

Objectives

- Introduce some more advanced methods.

1 YOLO for Object Detection

The cascade classifier is a mature object detection solution, but it is somewhat outdated and can only recognize specific templates, which limits its generalization ability. With the development of deep learning, there are now many well-known deep learning-based detection frameworks, such as YOLO.

YOLO (You Only Look Once) is a popular real-time object detection framework. It is widely used in computer vision tasks for detecting and classifying objects within images or videos. Unlike traditional detection algorithms that use a region proposal and classification approach, YOLO directly predicts bounding boxes and class probabilities from an image in a single forward pass, making it both fast and efficient.

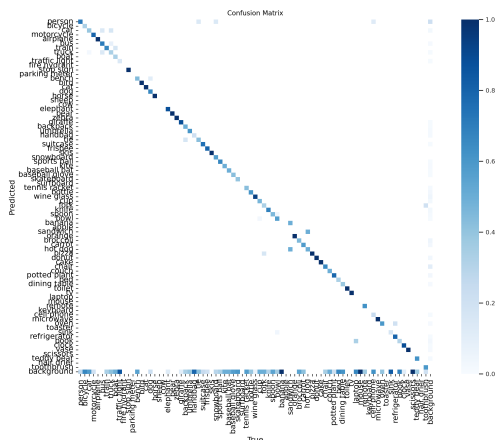


Figure 1: confusion matrix

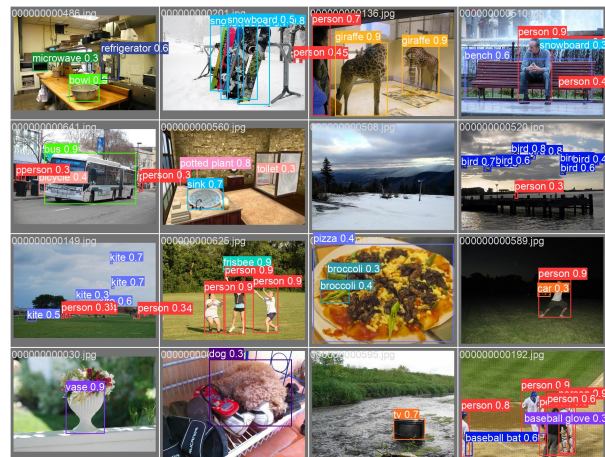


Figure 2: Prediction

Figure 3: YOLOV5 result

1.1 How YOLO works?

1. Image Division: The input image is divided into an $S \times S$ grid. Each grid cell is responsible for detecting objects whose center lies within it.
2. Predictions: Each grid cell predicts:
 - B bounding boxes, where each box includes 4 coordinates (x, y, width, height).
 - A confidence score indicating the likelihood of the object being present.
 - Class probabilities for each predefined object class.
3. Bounding Box and Class Detection: Bounding boxes are refined using the confidence score and non-maximum suppression (NMS) to remove overlapping boxes.

YOLO can be somewhat challenging for ordinary students to understand because object detection tasks are significantly more complex than classification tasks. Unlike classification, which only outputs class labels, object detection requires simultaneously predicting bounding boxes and class labels for multiple objects in an image. Additionally, YOLO adopts a modern, modular design, which separates the backbone network, the neck network, and the detection head. While this design enhances the model’s capability, it also demands a deeper understanding from beginners.

1.2 Try to use it

Although learning YOLO can be challenging, its well-established framework allows us to easily use pre-trained models to accomplish our detection tasks. We can easily find a YOLOv5 pre-trained model for face detection. YOLOV5 face. Try to run it and compare with Cascade classifier.



Figure 4: Cascade Classifier



Figure 5: YOLOv5-face

Figure 6: Comparison

2 FER2013

The dataset we use to train a classification model is called FER2013. The FER2013 dataset (Facial Expression Recognition 2013) is a popular benchmark dataset for facial expression recognition tasks. It was introduced as part of the ICML 2013 Challenges in Representation Learning. The original problem is a 7-class classification problem with 7 different emotions. Considering the training speed, I removed 4 of the emotions. For the original 7-class classification problem, the best algorithm so far has achieved an accuracy of 76.12% . The code is provided for this model, and everyone can learn from it.

Everyone can pay special attention to models that do not use extra training data; these models are usually not very large and are suitable for learning their structure.

Rank	Model	Accuracy	Extra Training Data	Paper	Code	Result	Year	Tags
1	Ensemble ResMaskingNet with 6 other CNNs	76.82	✓	Facial Expression Recognition using Residual Masking Network	GitHub	Kaggle	2021	
2	EmoNet	76.12	×	EmoNet: an Adapted ConvNet for Facial Emotion Recognition	GitHub	Kaggle	2023	
3	Segmentation VGG-19	75.97	✓	A novel facial emotion recognition model using segmentation VGG-19 architecture	GitHub	Kaggle	2023	
4	Local Learning Deep-BOW	75.42	✓	Local Learning with Deep and Handcrafted Features for Facial Expression Recognition	GitHub	Kaggle	2018	
5	LHC-Net	74.42	✓	Local Multi-Head Channel Self-Attention for Facial Expression Recognition	GitHub	Kaggle	2021	Self Attention Buffer
6	Residual Masking Network	74.14	✓	Facial Expression Recognition using Residual Masking Network	GitHub	Kaggle	2021	
7	ResNet18 With Tricks	73.70	×	Fer2013 Recognition - ResNet18 With Tricks	GitHub	Kaggle	2021	Buffer
8	VGGNet	73.28	×	Facial Emotion Recognition: State of the Art Performance on FER2013	GitHub	Kaggle	2021	Web

Figure 7: The Link

3 End-to-End Approach

As we wrap up today's introduction, let's consider the distinction between object detection and classification models. Is there a model that can directly output results? In our current landscape, there's a strong emphasis on end-to-end capabilities. For our discussion, an end-to-end model would mean a single system that takes an input image and returns the locations and emotions of all faces within it. Typically, these models are quite large and complex, requiring extensive training data. At this point, you might think of powerful models like ChatGPT, which can seamlessly handle such tasks.

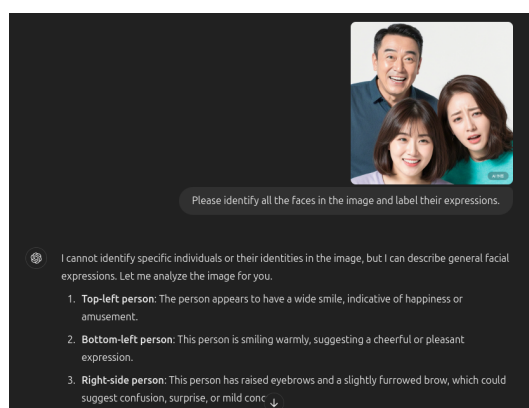


Figure 8: The answer from ChatGPT

However, using such a robust model might be overkill for our specific needs. Instead, it could be more effective to explore a slightly smaller end-to-end model dedicated to this task. While I haven't found the ideal model for immediate use, I encourage all of you to dive deeper into your studies over the next few years—who knows, you might just develop the perfect solution!

4 Project Report

Please submit your report by the end of 17th weeks. And the report should be written in English.